

it follows that $\ln f(x) = -\ln \sigma \sqrt{2\pi} - \frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2$

so that $\frac{\partial \ln f(x)}{\partial \mu} = \frac{1}{\sigma} \left(\frac{x-\mu}{\sigma} \right)$.

Hence, $E \left[\left(\frac{\partial \ln f(x)}{\partial \mu} \right)^2 \right] = \frac{1}{\sigma^2} E \left[\left(\frac{x-\mu}{\sigma} \right)^2 \right] = \frac{1}{\sigma^2} \cdot 1 = \frac{1}{\sigma^2}$

Thus, $\frac{1}{n E \left[\left(\frac{\partial \ln f(x)}{\partial \mu} \right)^2 \right]} = \frac{1}{n \frac{1}{\sigma^2}} = \frac{\sigma^2}{n}$.

Since $\bar{x}$ is unbiased and $\text{Var}(\bar{x}) = \frac{\sigma^2}{n}$, it follows that $\bar{x}$ is a minimum variance unbiased estimator of μ .

If V_0 be the variance of the most efficient estimator and V be the variance of any other estimator, then efficiency of the estimator is defined as: $\frac{V_0}{V} \leq 1$.

In sampling from a Normal distribution $N(\mu, \sigma^2)$, both the sample mean and variance are consistent estimators of μ . We know that the $\text{Var}(\bar{x}) = \frac{\sigma^2}{n}$ and

$\text{Var}(\text{Median}) = \pi \frac{\sigma^2}{2n}$. So, efficiency of Mean $= \frac{\frac{\sigma^2}{n}}{\pi \frac{\sigma^2}{2n}} = \frac{2}{\pi} = 0.64$. So, $\text{Var}(\bar{x})$ is

smaller than $\text{Var}(\text{Median})$. Hence, mean is more efficient than median in estimating the parameter μ . A statistic t which has the minimum variance among all estimators of θ is called **Minimum Variance (MV)** estimator.

30.3.4 Sufficiency

An estimator is said to be sufficient for a parameter, if it contains all the information in the sample regarding the parameter. More precisely, if $T_n = T(x_1, x_2, \dots, x_n)$ is an estimator of a parameter θ , based on a sample $x_1, x_2, \dots, x_n$ of size 'n' from the population with density $f(x, \theta)$ such that the conditional distribution of $x_1, x_2, \dots, x_n$ given T , is independent of θ , then T is a sufficient estimator for θ .

Factorization Theorem (Neyman)

It is very tedious to check whether a statistic is a sufficient estimator of a given parameter by following the above definition. It is easier to base it on the Factorization theorem which gives a necessary and sufficient condition for a statistic to be a sufficient estimator of a parameter.

Statement: $T = t(x)$ is sufficient estimator of the parameter θ if and only if the joint probability distribution or density function L of the random variable can be

expressed in the form $L = g_\theta[t(x)].h(x)$ where $g_\theta[t(x)]$ depends on θ and x only, through the values of $t(x)$ and $h(x)$ is independent of θ .

Note:

- i) By ‘a function independent of θ ’ we not only mean that it does not involve θ but also that its domain does not contain θ . For example, the function $f(x) = 1/2a; a - \theta < x < a + \theta$ and $-\infty < \theta < \infty$, depends on θ .
- ii) The original sample $X = (X_1, X_2, \dots, X_n)$, is always a sufficient statistic.
- iii) The most general form of the distributions admitting sufficient statistic is Koopman’s form and is given by $L = L(x, \theta) = g(x).h(\theta).\exp\{a(\theta) \psi(x)\}$ where $h(\theta)$ and $a(\theta)$ are functions of the parameter θ only and $g(x)$ and $\psi(x)$ are the functions of the sample observations only. The above equation represents the famous ‘exponential family of distributions’, of which most of the common distributions like the binomial, the Poisson and the normal with unknown mean and variance are its members.
- iv) Invariance property of sufficient estimator:

If T is a sufficient estimator for the parameter θ , and $\psi(t)$ is a one-to-one function of T , then $\psi(t)$ is sufficient for $\psi(\theta)$.

- v) Fisher – Neyman Criterion:

A statistic $t_1 = t_1(x_1, x_2, \dots, x_n)$ is sufficient estimator of parameter θ if and only if the likelihood function (joint p.d.f of the sample) can be expressed as:

$$L = \prod_{i=1}^n f(x_i, \theta) \\ = g_1(t_1, \theta). k(x_1, x_2, \dots, x_n)$$

where $g_1(t_1, \theta)$ is the p.d.f. of statistic t_1 and $k(x_1, x_2, \dots, x_n)$ is a function of sample observations only, independent of θ .

Note that this method requires working out the p.d.f (p.m.f) of the statistic $t_1(x_1, x_2, \dots, x_n)$, which is not always easy.

Check Your Progress 1

- 1) Define point estimation and interval estimation.

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- 2) State the characteristics of a good estimator.

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- 3) $x_1, x_2, \dots, x_n$ is a random sample from a normal population $N(\mu, 1)$. Show that $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i^2$, is an unbiased estimator of $\mu^2 + 1$.

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- 4) A random sample $(X_1, X_2, X_3, X_4, X_5)$ of size 5 is drawn from a normal population with unknown mean μ . Consider the following estimators to estimate μ :

i) $t_1 = (X_1 + X_2 + X_3 + X_4 + X_5) / 5$; and (ii) $t_2 = (2X_1 + X_2 + \lambda X_3) / 3$.

Find out λ . Are t_1 and t_2 unbiased? State the estimator which is best between t_1 and t_2 . Give reasons.

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- 5) Let $X_1, X_2, \dots, X_n$ be a random sample from a population with p.d.f $f(x, \theta) = \theta x^{\theta-1}$; $0 < x < 1$, $\theta > 0$. Show that $t_1 = \prod_{i=1}^n X_i$, is sufficient for θ .

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30.4 CRAMER-RAO INEQUALITY

To check whether a given unbiased estimator has the smallest possible variance, i.e., whether it is a minimum variance unbiased estimator (also called best unbiased estimator) then the variance of the unbiased estimator ($\hat{\theta}$) of parameter θ must satisfy the following inequality:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{nE\left[\left(\frac{\partial \ln f(x)}{\partial \theta}\right)^2\right]}$$

where, $f(x)$ is the value of population density at x and n the sample size. This is the Cramer Inequality which leads us to the following result:

If $(\hat{\theta})$ is an unbiased estimator of θ and

$$\text{Var}(\hat{\theta}) = \frac{1}{nE\left[\left(\frac{\partial \ln f(x)}{\partial \theta}\right)^2\right]},$$

Then $(\hat{\theta})$ is an unbiased estimator of θ . Here, the term in the denominator captures the **information** $I(\theta)$ about θ that is provided by the sample. Thus, smaller the variance indicates greater information.

The Cramer-Rao inequality holds given the following assumptions, which are known as the ‘regularity conditions for Cramer-Rao inequality’.

- i) The parameter space Θ is a non-degenerate open interval on the real line R^1 $(-\infty, \infty)$.
- ii) For almost all $x = (x_1, x_2, \dots, x_n)$, and for all $\theta \in \Theta$, $\frac{\partial}{\partial \theta} f(x, \theta)$ exists, the exceptional set, if any, is independent of θ .
- iii) The range of integration is independent of the parameter θ , so that $f(x, \theta)$ is differentiable under integral sign.
- iv) The conditions of uniform convergence of integrals are satisfied so that differentiation under the integral sign is valid.
- v) $L(\theta) = E \left[\left\{ \frac{\partial}{\partial \theta} \log f(x, \theta) \right\}^2 \right]$ exists and is positive for all $\theta \in \Theta$.

An unbiased estimator t of $\gamma(\theta)$ for which Cramer-Rao lower bound is attained is called a minimum variance bound (MVB) estimator.

30.4.1 MVUE and Blackwellisation

Cramer-Rao inequality provides us a technique of finding if the unbiased estimator is also a MVUE or not. Here, since the regularity conditions are very strict, its application becomes restrictive. Moreover, MVB estimator is not the same as MVUE since the Cramer-Rao lower bound may not always be satisfied. Moreover, if the regularity conditions are violated, then the least attainable variance may be less than the Cramer-Rao bound. There is a method of obtaining MVUE from an unbiased estimator using sufficient statistic. This technique is called Blackwellisation named after D. Blackwell. The result is contained in the following theorem due to C. R. Rao and D. Blackwell.

30.4.2 Rao-Blackwell Theorem

Let X and Y be random variables such that $E(Y) = \mu$ and $\text{var}(Y) = \sigma_Y^2 > 0$. Now, if $E(Y | X = x) = \phi(x)$, then i) $E[\phi(x)] = \mu$ and (ii) $\text{var}[\phi(x)] \leq \text{var}(Y)$.

The above result, called the Rao-Blackwell theorem, enables us to obtain the MVUE through a sufficient statistic. If a sufficient estimator exists for a parameter, then in our search for MVUE we may restrict ourselves to functions of the sufficient statistic.

The above theorem can be stated in a slightly different manner as follows:

Let $U = U(x_1, x_2, \dots, x_n)$ be an unbiased estimator of parameter $\gamma(\theta)$ and let $T = T(x_1, x_2, \dots, x_n)$ be a sufficient statistic for $\gamma(\theta)$. Consider the function $\phi(T)$ of the sufficient statistic defined as $\phi(T) = E(Y | T = t)$ which is independent of θ (since T is sufficient for $\gamma(\theta)$). Then $E\phi(T) = \gamma(\theta)$ and $\text{var } \phi(T) \leq \text{var } (U)$.

This result implies that starting with an unbiased estimator U , we can improve upon it by defining a function $\phi(T)$ of the sufficient statistic given as $\phi(T) = E(Y | T = t)$. This technique of obtaining improved estimator is called Blackwellisation. If in addition, the sufficient statistic T is also complete, then the estimator $\phi(T)$ discussed above will not only be an improved estimator over U but also the ‘best (unique)’ estimator.

30.5 ASYMPTOTIC PROPERTIES

The section focuses on the study of statistics with respect to the asymptotic theory or the large sample theory for understanding the properties of the estimators and statistical tests. This essentially amounts to discussing the properties of estimators that hold true when the sample size in consideration becomes large i.e., the properties of estimators are studied for the limit $n \rightarrow \infty$. This section enumerates such properties.

30.5.1 Unbiasedness

We refer to $\hat{\theta}_n$ as an asymptotically unbiased estimator of θ if its expected value converges to θ when sample becomes very large i.e., as $n \rightarrow \infty$. The following expression depicts this property of the estimators: $\lim_{n \rightarrow \infty} E(\hat{\theta}_n) = \theta$.

If this does not hold true, then we refer to the estimators as being asymptotically biased.

30.5.2 Consistency

$\hat{\theta}$ converges in probability to θ if for any $\varepsilon > 0$, we can write the following:

$$\lim_{n \rightarrow \infty} \Pr(|\hat{\theta} - \theta| > \varepsilon) = 0$$

If this holds true, then we can also say that $\hat{\theta}$ is consistent for θ . Intuitively, this means that for a reasonably large sample size the sample statistic $\hat{\theta}$ equals θ . Thus, when the sample size n is large we can be practically be certain that an error made with a consistent estimator will be less than any small pre-assigned positive constant. This kind of convergence expressed in the form of limit, as in the above equation, is generally called ‘convergence in probability’. In practice, we can ensure whether an estimator is consistent or not by using the following sufficient condition, which is a corollary of Chebyshev’s theorem.

If $(\hat{\theta})$ is an unbiased estimator of θ and $(\hat{\theta}) \rightarrow 0$ as $n \rightarrow \infty$, then $(\hat{\theta})$ is a consistent estimator of θ .

Example: Show that for a random sample from a normal population the sample variance S^2 is a consistent estimator of σ^2 .

If sample variance of a random variable is S^2 from an infinite population with a finite variance σ^2 then we can show $E(S^2) = \sigma^2$.

$$\begin{aligned} E(S^2) &= E\left[\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2\right] \\ E(S^2) &= E\left[\frac{1}{n-1} \sum_{i=1}^n \{(X_i - \mu) + (\mu - \bar{X})\}^2\right] \\ &= \frac{1}{n-1} \left[\sum_{i=1}^n E\{(X_i - \mu)^2\} - nE\{(\mu - \bar{X})^2\} \right] \\ E(S^2) &= \frac{1}{n-1} \left[\sum \sigma^2 - n \frac{\sigma^2}{n} \right] = \sigma^2 \end{aligned}$$

Note: The above result is owing to the fact that $E\{(X_i - \mu)^2\} = \sigma^2$ and $E\{(\mu - \bar{X})^2\} = \sigma^2 / n$

Keep it in mind that S^2 is an unbiased estimator of the variance (σ^2) of the infinite population. It is not an unbiased estimator of the variances of the finite population and in neither case is S an unbiased estimator of σ .

Since S^2 is an unbiased estimator of σ^2 , it remains to be shown that $VarS^2 \rightarrow 0$ as $n \rightarrow \infty$. We find that for a random sample from a normal population, $VarS^2 = \frac{2\sigma^4}{n-1}$. Hence, it follows that $VarS^2 \rightarrow 0$ as $n \rightarrow \infty$. Thus, the sample variance S^2 is a consistent estimator of the variance of a normal population σ^2 .

30.5.3 Convergence in Distribution and Asymptotic Normality

Convergence in Distribution: A sequence of random variables Z_n converges in distribution to a continuous random variable Z if for any real z , the following holds true:

$$\Pr(Z_n \leq z) \rightarrow \Pr(Z \leq z) \text{ as } n \rightarrow \infty$$

This essentially means that the shape of the distribution of Z_n gets closer and closer to that of Z .

Asymptotic Normality: For this particular property, we say that if the limiting distribution of Z_n is normally distributed i.e., $Z \sim N(\mu, \sigma^2)$, then Z_n is said to have asymptotic normal distribution. If $Z \sim N(0, 1)$, then Z_n has an asymptotic standard normal distribution.

Therefore, for the case when n is large, we have the following approximation: $\Pr(Z_n \leq z) \approx \Phi(z)$ where $\Phi(z)$ is the standard normal c.d.f.

Central Limit Theorem: To explain this, let us suppose that $X_1, X_2, \dots, X_n$ form an independent and identically distributed sample from a distribution with

$E(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$. Then, we say that, $Z_n = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1)$ where $\xrightarrow{d}$ denotes ‘convergence in distribution’.

Intuitively, this means that if we have a population with mean μ and standard deviation σ and also take large random samples (usually $n \geq 30$) from the population with replacement, then the distribution of the sample means will be approximately normally distributed. Additionally, the mean of samples will be approximately equal to mean of the whole population.

30.6 METHODS OF ESTIMATION

When we are given a parameter, for example population proportion (p) or population mean (μ), the ultimate objective is to arrive at a number that best represents the true value of the parameter. To achieve the objective, a sample is used to compute that number which is a good guess for this true value of the parameter. In this section, we will focus on three methods of estimation, namely: Maximum Likelihood, Generalized Method of Moments and Partial Least Squares.

30.6.1 Maximum Likelihood (ML)

Maximum Likelihood is the most widely used method of constructing an estimator for an unknown parameter θ . In case the sample is large, it gives an excellent estimator of the unknown parameter.

To understand the concept, let us suppose that the random variable $X_1, \dots, X_n$ is a random sample from a distribution $f(x|\theta)$. Further, for every observed random sample of the form $x_1, \dots, x_n$, we write $f(x_1, \dots, x_n|\theta) = f(x_1|\theta) \cdot \dots \cdot f(x_n|\theta)$. This function is denoted as $L(\theta)$ and is called the likelihood function. It is a joint density function if $f(x|\theta)$ is a p.d.f. or the joint probability function if $f(x|\theta)$ is a p.m.f.

Intuitively, for any observed data of $x_1, \dots, x_n$, we have to consider a value of θ for which the likelihood function $L(\theta)$ is maximum. Therefore, that value of θ will serve as the estimate of the unknown parameter θ . It essentially means that we are choosing the parameter that yields the likelihood of having the data that has been obtained as maximum.

In case we had no numerical values attributed to the random variables $X_1, \dots, X_n$, ML (Maximum Likelihood) function will yield us a function of n variables which itself will be referred to as the maximum likelihood estimate $\hat{\theta}$. However, in case actual data of numerical values are available, the estimate will have a

particular numerical value. It is then referred to as the ‘maximum likelihood estimator’.

Given the form of the likelihood function, it is not an easy task to maximise it. Therefore, maximising this function is equivalent to maximizing $\log L(\theta)$ as $\log$ is a monotonically increasing function. So, we get:

$$l(\theta) = \log L(\theta) = \log \prod_{i=1}^n f(X_i|\theta) = \sum_{i=1}^n \log f(X_i|\theta)$$

Hence, maximizing $l(\theta)$ with respect to θ will yield the Maximum Likelihood Estimator (MLE).

30.6.2 Generalized Method of Moments (GMM)

This method of estimation is in many ways similar to the Maximum Likelihood (ML) method of estimation. The GMM puts to use the assumptions about specific moments of the random variables rather than making assumptions about the entire distribution. Therefore, this approach makes the GMM estimators more robust than the estimators calculated by the method of maximum likelihood.

To describe it further, suppose the data in consideration is composed of T observations $\{Y_t\}_{t=1, \dots, T}$, where Y_t is an n -dimensional multivariate random variable. As a condition to use GMM, we require the moment conditions i.e. a vector-valued function $g(Y, \theta)$ for which $m(\theta_0) \equiv E[g(Y_t, \theta)] = 0$ where E signifies the expectation and Y_t is a generic observation. Intuitively, applying GMM amounts to replacing the expected value $E[\cdot]$ with the sample average:

$$\hat{m}(\theta) \equiv \frac{1}{T} \sum_{t=1}^T g(Y_t, \theta)$$

The next step is to minimise the norm of the expression so that we get to our estimator θ_0 . Such estimators, called the GMM estimators, are known to be asymptotically normal, consistent and the most efficient in the class of estimators that do not use/require any information other than the one gathered from the moment conditions.

30.6.3 Partial Least Squares (PLS)

Partial Least Squares (PLS) method is used for estimation when either the number of explanatory (independent) variables are quite large as compared to the number of observations in the data or these explanatory variables are correlated. This method of estimation decreases the number of independent variables so that we just have relevant variables for the analysis. It further uses another method known as Principal Component Analysis to identify the uncorrelated components. Principal component analysis (PCA) is a method that reduces the dimensionality of large data sets by transforming the large set of variables into a smaller one that still contains most of the information in the original large set of variables.

Check Your Progress 2

- 1) Enumerate and discuss the asymptotic properties of estimators.
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- 2) Explain the concept of Central Limit Theorem.
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- 3) Suppose you have a coin that you think is biased. You flip it 3 times and get the sequence HHT. What is the maximum likelihood estimate (MLE) for the probability of getting heads?
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- 4) Suppose $x_1, \dots, x_n$ are independent and identically distributed samples from a distribution with density

$$f_x(x|\theta) = \begin{cases} \frac{\theta x^{\theta-1}}{2^\theta}, & 0 \leq x \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

Find the MLE for θ .

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30.7 LET US SUM UP

We began the unit with the characteristics of good estimators and then we proceeded to the topic of Cramer Rao Inequality. We then discussed the properties of estimators that hold true when the sample size becomes quite large i.e., the asymptotic properties of the estimators like unbiasedness, consistency, convergence in distribution and asymptotic normality. Lastly, the methods of estimation were discussed to understand the ways of computing the best value of the parameter using the methods of Maximum Likelihood, Generalized Method of Moments and Partial Least Squares.

30.8 KEY WORDS

Central Limit Theorem : Intuitively, this means that if we have a population with mean μ and standard deviation σ and also take large random samples (usually $n \geq 30$) from the population with replacement, then the distribution of the sample means will be approximately normally distributed.

Completeness : This is the property of an estimator which ensures that the distributions corresponding to different estimates are distinct. Hence, it is possible for a 'complete statistic' to provide no information about θ . It is therefore, also necessary for it to be 'sufficient'.

Cramer-Rao Inequality : If t is an unbiased estimator of $\gamma(\theta)$, a function of parameter θ , then

$$\text{var}(t) \geq \frac{\left[\frac{\partial}{\partial \theta} L(x, \theta) \right]^2}{E \left[\frac{\partial}{\partial \theta} \log L \right]} = \frac{[\gamma'(\theta)]^2}{I(\theta)}$$

$$\text{and } I(\theta) = E \left[\left\{ \frac{\partial}{\partial \theta} \log L(x, \theta) \right\}^2 \right],$$

where $I(\theta)$ is the information on θ , supplied by the sample. In other words, Cramer-Rao inequality provides a lower bound $\frac{[\gamma'(\theta)]^2}{I(\theta)}$ to the variance of an unbiased estimator of $\gamma(\theta)$.

Consistency : T_n is a consistent estimator of $\gamma(\theta)$ if for every $\varepsilon > 0$, $\eta > 0$, there exists a positive integer $n \geq m(\varepsilon, \eta)$ such that $P[|T_n - \gamma(\theta)| < \varepsilon] \rightarrow 1$ as $n \rightarrow \infty \Rightarrow P[|T_n - \gamma(\theta)| < \varepsilon] > 1 - \eta$; $\forall n \geq m$ where m is some very large value of n .

Efficiency : T is the best estimator (or efficient estimator) of $\gamma(\theta)$ if it is consistent and normally distributed and if $\text{avar}(T) \leq \text{avar}(T')$ whatever the other consistent and asymptotically normal estimator T' may be.

Generalized Method of Moments : In case of GMM, it puts to use the assumptions about specific moments of the random variables rather than using assumptions about the entire distribution. Therefore, this property makes the GMM estimators more robust than the estimators calculated by the method of maximum likelihood.